

Configuring VPC Peering Between Two VPCs

To Begin with the Lab

Summary of the Lab

This lab demonstrated how to configure VPC Peering between two VPCs located in different AWS Regions (Mumbai and US East–Ohio) to enable private communication. Separate VPCs with non-overlapping CIDR blocks, subnets, and EC2 instances were created without public IPs. Initial connectivity between the instances failed because the VPCs were isolated. A VPC Peering Connection was then created, accepted, and activated, followed by updating the route tables in both VPCs to route traffic through the peering connection. After the configuration, the EC2 instances successfully communicated using their private IP addresses over the AWS network without requiring an Internet Gateway or public internet access.

Note: This lab demonstrates communication between two VPCs located in different AWS Regions. Both VPCs are created in the same AWS account, but the same process also works for VPCs in different AWS accounts.

Open the **AWS Management Console** and switch to the **Mumbai (ap-south-1)** region. Create a new VPC named **Ind-VPC** with the IPv4 CIDR block **192.168.0.0/24**, ensuring that its address range is unique.

The screenshot shows the AWS Management Console interface for creating a new VPC. The top navigation bar indicates the region is 'Asia Pacific (Mumbai)'. The breadcrumb trail shows 'VPC > Your VPCs > Create VPC'. The main heading is 'Create VPC' with an 'Info' link. Below this, a description states: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.' The 'VPC settings' section contains the following fields:

- Resources to create** (Info link): A note says 'Create only the VPC resource or the VPC and other networking resources.' There are two radio buttons: 'VPC only' (selected) and 'VPC and more'.
- Name tag - optional** (Info link): A note says 'Creates a tag with a key of 'Name' and a value that you specify.' The text input field contains 'Ind-vpc'.
- IPv4 CIDR block** (Info link): Two radio buttons are present: 'IPv4 CIDR manual input' (selected) and 'IPAM-allocated IPv4 CIDR block'.
- IPv4 CIDR**: A text input field contains '192.168.0.0/24'. A small note below the field states: 'CIDR block size must be between /16 and /28.'

Create a subnet inside **India-VPC** by selecting the VPC, assigning a name such as **India-Subnet**, choosing the **ap-south-1a** Availability Zone, and using the entire CIDR block **192.168.0.0/24**. Since no Internet Gateway is configured, this subnet functions as a private subnet.

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Ind-Subnet
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/24

IPv4 subnet CIDR block
192.168.0.0/24 256 IPs

Launch an EC2 instance named **India-Server** inside the India subnet. Disable **Auto-assign Public IP** and configure the Security Group to allow **All Traffic** so that connectivity testing between the VPCs is not blocked by firewall rules.

i-08a1b400006d0ddc6 (Ind-Server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

VPC ID

vpc-0906ca639c7126c2d (Ind-vpc)

Availability zone ID

aps1-az1

▼ IP addresses

info

Subnet ID

subnet-0b3e2d0f3b340d406 (Ind-Subnet)

Outpost ID

-

Availability zone

ap-south-1a

Public IPv4 address

-

Private IPv4 addresses

192.168.0.181

IPv6 addresses

-

Secondary private IPv4 addresses

-

Carrier IP addresses (ephemeral)

-

Security Group inbound rule for US-server

Inbound rules (2)						
Search Manage tags Edit inbound rules						
☐	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-094ea27083dddc90f	IPv4	All traffic	All	All
☐	-	sgr-00d3b04b14355c09e	IPv4	SSH	TCP	22

Switch to the **US East (Ohio)** region and create another VPC named **US-VPC** using a different CIDR block such as **192.168.1.0/24**. The CIDR ranges of both VPCs must be different because overlapping CIDR blocks are not supported for VPC Peering.

☰ VPC > Your VPCs > Create VPC 🔍 ⚙️ 📄

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

US-vpc

IPv4 CIDR block Info
☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
192.168.1.0/24
CIDR block size must be between /16 and /28.

Create a subnet inside **US-VPC**, assign it a name such as **US-Subnet**, choose the appropriate Availability Zone, and use the VPC CIDR block **192.168.1.0/24**.

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
US-Subnet
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1a

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.1.0/24

IPv4 subnet CIDR block
192.168.1.0/24 256 IPs

Launch another EC2 instance named **US-Server** inside the US subnet. Disable **Auto-assign Public IP** and configure the Security Group to allow **All Traffic**, just as you did for the India server.

Network settings

VPC - required Info
vpc-01f4bad02398b15fb (US-vpc)
192.168.1.0/24

Subnet Info
subnet-0e84e0b03b3a8ae57 **US-Subnet** [Create new subnet](#)
VPC: vpc-01f4bad02398b15fb Owner: 463646775279
Availability Zone: us-east-1a (us-east-1a) Zone type: Availability Zone
IP addresses available: 251 CIDR: 192.168.1.0/24

Auto-assign public IP Info
Disable

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☒ Create security group ☐ Select existing security group

Security group name - required
US-Server
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./[!@#%&()*~`]{1,255}

Description - required Info
launch-wizard-31 created 2026-06-30T11:15:57.492Z

Summary

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...[read more](#)
ami-06067086cf86c58e6

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Security Group inbound rule for US-server

Inbound rules (2)							
Q Search							
☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	
☐	-	sgr-01e248117750ba51a	IPv4	SSH	TCP	22	
☐	-	sgr-07f4c7b80432fa830	IPv4	All traffic	All	All	

Since both EC2 instances are located in private subnets, create an **EC2 Instance Connect Endpoint** for the India VPC. After the endpoint becomes **Available**, use it to connect to the India EC2 instance through the AWS Console.

VPC > Endpoints > Create endpoint

Network settings
Select the VPC in which to create the endpoint

VPC
Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

vpc-0906ca639c7126c2d (Ind-vpc)

Additional settings

Security groups (1/2) info

Q Search

VPC ID: vpc-0906ca639c7126c2d X Clear filters

☐	Group ID	Group name	VPC ID	Description
☐	sg-05ea8edacc8e06972	Ind-server	vpc-0906ca639c7126c2d	launch-wizard-1181 created 2026-06-30T11:05:12.883Z
☒	sg-08d833a11ab452101	default	vpc-0906ca639c7126c2d	default VPC security group

sg-08d833a11ab452101 X

Subnet
Select the Subnet in which to create the endpoint

Subnet
Select the subnets in which to create the endpoint.

subnet-0b3e2d0f3b340d406 (Ind-Subnet)

From the India EC2 instance, attempt to ping the **Private IP Address** of the US EC2 instance. The ping request will fail because, by default, separate VPCs cannot communicate with each other, even if they belong to the same AWS account.

EC2 > Instances > i-08a1b40006d0ddc6 > Connect to instance

Connection type

☐ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☒ **Connect using a Private IP**
Connect using a private IP address and a VPC endpoint

Private IPv4 address
192.168.0.181

IPv6 address
-

EC2 Instance Connect Endpoint
Only endpoints that have completed the creation process can be selected. The process can take up to 15 minutes. If you create an endpoint, refresh this list to check if its in the available state.

eice-0dc915397811e5d52 (create-complete)

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

Max tunnel duration (seconds)
The maximum allowed duration of the SSH connection. Must comply with the maxTunnelDuration condition (if specified) in the IAM policy.

3600

Min 1 second. Max 3600 seconds (1 hour).

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Now after the connecting to the instance, we will try to ping the private IP address of US-Server

```
[ec2-user@ip-192-168-0-181 ~]$ ping 192.168.1.219
PING 192.168.1.219 (192.168.1.219) 56(84) bytes of data.
```

Return to the **Mumbai** region and open the **Peering Connections** section in the VPC console. Click **Create Peering Connection**, provide a name such as **Ind-US**, select **India-VPC** as the requester VPC,

Peering connections info

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Acceptor VPC
No peering connection found				

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately.

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
IND-US

Select a local VPC to peer with

VPC ID (Requester)
vpc-0906ca639c7126c2d (Ind-vpc)

VPC CIDRs for vpc-0906ca639c7126c2d (Ind-vpc)

CIDR	Status	Status reason
192.168.0.0/24	Associated	-

choose the **US East (Orio)** region, and select **US-VPC** as the acceptor VPC.

Copy the VPC Id from the US region where we have created vpc.

Select another VPC to peer with

Account
☒ My account
☐ Another account

Region
☐ This Region (ap-south-1)
☒ Another Region
United States (Ohio) (us-east-2)

VPC ID (Acceptor)
vpc-0957d3dca45323b02

Click **Create Peering Connection**

Switch to the **US East (Ohio)** region, open **Peering Connections**, select the pending request, and click **Accept Request**.

Peering connections (1/1) info

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Acceptor VPC
-	pcx-00c7c36c15478a8c4	Pending acceptance	vpc-0906ca639c7126c2d	vpc-0957d3dca45323b02 / US

Actions menu:
View details
Accept request
Reject request
Edit DNS settings
Manage tags
Delete peering connection

Wait until the peering connection status changes to **Active** in both regions.

Peering connections (1) info

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Acceptor VPC
-	pcx-00c7c36c15478a8c4	Active	vpc-0906ca639c7126c2d	vpc-0957d3dca45323b02 / US

Open the **Route Table** associated with **India-VPC** and edit its routes. Add a new route with the destination **192.168.1.0/24** and select the newly created **VPC Peering Connection** as the target, then save the changes.

Destination: 192.168.0.0/24

Target: local

Status: Active

Propagated: No

Route Origin: CreateRouteTable

Destination: 192.168.1.0/24

Target: Peering Connection

Status: -

Propagated: No

Route Origin: CreateRoute

Remove

Add route

Cancel Preview Save changes

Switch to the **US East (Ohio)** region and edit the **US-VPC Route Table**. Add another route with the destination **192.168.0.0/24** and choose the same **VPC Peering Connection** as the target, then save the route.

Destination: 192.168.1.0/24

Target: local

Status: Active

Propagated: No

Route Origin: CreateRouteTable

Destination: 192.168.0.0/24

Target: Peering Connection

Status: Active

Propagated: No

Route Origin: CreateRoute

Remove

Add route

Cancel Preview Save changes

Verify that communication is occurring entirely through **Private IP Addresses** over the AWS network. No Internet Gateway or Public IP Address is required for communication between the two VPCs after peering has been configured.

```
[ec2-user@ip-192-168-0-181 ~]$ ping 192.168.1.219
PING 192.168.1.219 (192.168.1.219) 56(84) bytes of data:
64 bytes from 192.168.1.219: icmp_seq=1 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=2 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=3 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=4 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=5 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=6 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=7 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=8 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=9 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=10 ttl=127 time=215 ms
64 bytes from 192.168.1.219: icmp_seq=11 ttl=127 time=214 ms
64 bytes from 192.168.1.219: icmp_seq=12 ttl=127 time=214 ms
```